Data Trusted Sharing Delivery: A Blockchain-Assisted Software-Defined Content Delivery Network

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***Abstract*—The 6G wireless network aims to forge a new spectrum, high technical standards of high time and phase synchronization accuracy, and 100% geographic coverage to connect trillions of devices flexibly and efficiently in the future. However, as connectivity increases and applications become novel, it is a challenge to ensure the privacy and security of networks and applications. Blockchain is seen as a promising technology that can improve efficiency, reduce costs, mitigate security, and privacy threats, and establish a trusted data-sharing environment. This article presents a trusted framework based on blockchain technology from the perspective of how to build a trusted software-defined content delivery network. As the peer node of the blockchain, the software-defined network (SDN) controller establishes trust between different regions and a wide range of participants, realizing peer autonomy and flexible business orchestration. The two main purposes of the architecture are to enhance the security of network communications and establish trust relationships between entities in different domains. It includes trusted communication based on routing sandbox, service choreography based on blockchain, proxy server selection strategy based on model predictive control (MPC), and optimization consensus based on practical Byzantine fault tolerance. Some simulation experiments verify the effectiveness of the theoretical method.**

***Index Terms*—Blockchain, routing sandbox, sixth-generation (6G), software-defined network (SDN).**

I. Introduction

As THE standardization of 5G communication has been completed, the vision and planning for sixth-generation (6G) communication have begun. 6G aims to cast a new spectrum, high technical standards with high time and phase synchronization accuracy, and 100% geographic coverage to connect trillions of devices flexibly and efficiently in the future [1]. This will involve a large number of heterogeneous devices with limited resources interacting with our physical world. As a result, security and privacy issues stand in the way of further deployment and adoption of 5/6G [2]. From the perspective of establishing trusted software-defined content delivery, we will try to provide blockchain-based solutions to trust and privacy issues encountered during the transition from 4G to 5/6G networks.

With the development of 5/6G, content delivery networks (CDNs) have become an important way to alleviate Internet congestion and improve users' service experience [3]. CDN improves users' Quality of Service (QoS) by backing up the content of the origin server to a replica server that is closer to users. Thus, the user's requests are redirected to the replica server according to the user's geographical location [4]. However, CDN cannot perceive network topology and cannot obtain global network status information, so it cannot select the optimal server for users from a global perspective. Specifically, the replica server with the nearest geographic location is not the best choice for redirection from a global perspective [5].

To cope with this problem, a software-defined network (SDN) with more advantages in security, reliability, and maintainability is a promising solution. In the SDN architecture, network control is separated from the data plane. Usually, there is a controller (or cluster) which is responsible for collecting the topology and flow of the entire network, calculating the forwarding path, and sending the flow table to the switch. The switch only performs forwarding actions according to the flow table rules [6]. In addition, SDN has programmability. It can dynamically allocate underlying network resources through programming. By assuming a part of the CDN's function of obtaining network information, it not only reduces the CDN load but also strengthens the control of the underlying network [7], [8]. Therefore, the software-defined CDN can make up for the shortcomings of CDN that cannot obtain complete and accurate network status information.

However, software-defined CDNs still face many security challenges in practice. When SDN controllers are deployed in a centralized manner, it is easy to cause single points of failure, such as controller hijacking, DDoS attacks, etc., while distributed deployment controllers are more likely to encounter Internal attacks, such as malicious node intrusion [9] and consensus issues [10]. The application of blockchain technology to ensure the security of the data plane and control plane in SDN has received great attention from the academic community,

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because the blockchain can provide secure and trusted interactions in distributed peer-to-peer networks [11]. The blockchain uses asymmetric encryption technology to verify every new transaction in the control plane. In this case, the blockchain allows unknown entities to exchange data with each other without a trusted third party [12]. Therefore, blockchain not only enhances the security of distributed controllers and various forwarding devices but also improves the robustness of the software-defined CDN.

This article mainly focuses on the application of SDN to provide a global view for the selection of the optimal replica server in the CDN and further build a consortium blockchain on the SDN control plane to respond to security threats. The specific contributions of this article are summarized as follows.

1) A software-defined CDN architecture based on blockchain is proposed. On the one hand, it perceives the network topology through the SDN control plane and then provides a global view for the optimal selection of CDN replica servers. On the other hand, the consortium blockchain is built on the SDN control plane where distributed controllers participate in the consensus process in order to deal with single points of failure, malicious node intrusion, and consensus issues.
2) Model predictive control (MPC) is used to solve the problems of joint optimization delay and bandwidth satisfaction in the process of selecting the optimal replica server in the software-defined CDN. The importance of the two optimization goals of response time and bandwidth satisfaction can be balanced by adjusting the weight value according to actual network requirements.
3) In order to resist malicious nodes intrusion and improve reliability and security, not only the consortium blockchain is used to verify the identity of the controller but also a fuzzy synthetic evaluation model is established to evaluate the trust of the SDN controller to prevent untrusted nodes from joining.
4) In order to overcome the shortcomings of low consensus efficiency and poor self-healing when the practical Byzantine fault tolerance (PBFT) consensus algorithm is directly applied in the SDN scenario, we propose a simplified and PBFT consensus mechanism based on trust evaluation, which is referred to as trusted simplified PBFT (TS-PBFT).

The rest of the organization structure is as follows. Section II reviews the related work. Section III presents the blockchain-assisted software-defined CDN architecture. The blockchain consensus mechanism is introduced in Section IV. Section V describes the path compute element to optimal CDN surrogate. The experimental results and corresponding discussions are described in Section VI. Section VII gives the conclusion and some problems to be solved in the future.

II. Related Work

In this section, we review some interesting research on software-defined CDNs, blockchain, 5/6G, and Internet of Things (IoT) security.

A. Software-Defined Content Delivery Network

Tran *et al.* [13] proposed an SDN-based intelligent CDN architecture, where the controller makes cross-domain decisions in a centralized manner, and the MABRESE algorithm is applied to solve the server selection problem. Fu *et al.* [14] used the programmability and virtuality of SDN/NFV to design a scalable CDN architecture, where load-based redirection algorithms are applied to accurately and fine-grained control of load management traffic routing methods. In [15], CDN providers use OpenFlow switches to manage the underlying infrastructure. It also uses SDN controllers to achieve a global view of CDNs, thereby improving traditional load-balancing algorithms to make better decisions.

A simple and effective CDN request rerouting architecture based on SDN is proposed in [16], which can achieve user–server allocation with higher flexibility and requires fewer additional components to achieve the same rerouting performance. Mowla *et al.* [17] proposed a cognitive detection and defense mechanism in the SDN-based CDNi architecture, which can more accurately detect and defend DDoS attacks. Sharma *et al.* [18] proposed a distributed and secure SDN architecture (DistBlockNet) using blockchain technology. In this network, unconfident members can interact with each other without a trusted third party. The blockchain-based SDN architecture can inherit many advantages of SDN and blockchain, including scalability, reliability, and security [19]. In order to ensure the security of SDN on different planes and related interconnections, Aujla *et al.* [20] proposed BlockSDN, a framework that uses blockchain as an SDN network service, which can deal with malware intrusion on the data plane and DDoS attacks on the control plane. Aujla *et al.* [20] discussed the feasibility of integrating blockchain and SDN in IoT and proposed a blockchain-based SDN controller architecture. This architecture uses public and private blockchains for P2P communication between IoT devices and SDN controllers, with higher throughput, lower latency, and lower energy consumption.

B. Blockchain-Enabled Privacy Protection in IoT

As mentioned in [21], “Blockchain has been regarded as a promising technology for IoT, since it provides significant solutions for decentralized networks that can address trust and security concerns, high maintenance cost problems, and so on.” The blockchain consensus mechanism allows peer-to-peer transactions to take place in a distributed manner without third parties. By integrating a public chain running the delegated PoS (DPOS) consensus and subchains running the PBFT consensus, [22] achieve higher transaction throughput, less execution time, diverse privacy protection, and access control compared with traditional POW/POS-based blockchain. Furthermore, Zou *et al.* [23] considered the interference behavior of Byzantine elastic protocol in wireless networks, which is a very critical and realistic behavior that must be considered in modern wireless networks. Compared with PoW and Proof of Stake (PoS) that have been widely used in blockchain, direct acyclic graph (DAG) consensus proposed in [24] and [25] can overcome some weaknesses, such as high resource consumption, high transaction fee, low transaction throughput, and long confirmation delay.

In order to better combine blockchain and IoT, Zheng and Cai [26] investigated the problem of data sharing for IoT systems, where multiple data consumers exist in different stages. In the network topology, Yu *et al.* [27] defined the dynamicity in terms of localized topological changes in the vicinity of each node, rather than a global view of the whole network. A local dynamic model suits distributed algorithms better than a global one. In terms of computing power, Cao *et al.* [28] proposed multiaccess edge computing as a supplement to the traditional remote cloud center, which is deployed in the area adjacent to the device side of the IoT and is also considered as a promising 5G heterogeneous network technology. In terms of wireless communication, Xu *et al.* [29] presented wChain, a blockchain protocol specifically designed for multihop wireless networks that deeply integrates wireless communication properties and blockchain technologies under the realistic SINR model. In terms of physical, the concept of physical-layer anonymity is introduced in [30], which reveals that the receiver can reveal the identity of the sender only by analyzing the physical-layer information, namely, signaling mode and channel characteristics. Furthermore, Wei *et al.* [31] overviewed a distributed approach that utilizes interference in power-constrained IoT devices and applications to provide efficient, hardware efficient secure transmission for IoT downlinks. Cai and Zheng [32] considered conservation of energy and privacy protection to implement an efficient data uplinks scheme by introducing an acceptable amount of additional content. There is also some promising research on data trust access and AI. In [33], a blockchain-enhanced security access control scheme that supports traceability and revocability has been proposed in IoT for smart factories, which achieved secure storage, access control, information update, and deletion for smart factory data. Tan *et al.* [34] proposed a HoneyNet approach that includes both threat detection and situational awareness to enhance the security and resilience of Artificial Intelligence of Things (AIoT). It is more suitable for intelligent IoT than traditional network security methods.

C. Blockchain Meets 5/6G

As a result in [35], 6G wireless networks improve on 5G by further increasing reliability, speeding up the networks and increasing the available bandwidth. However, along with the enhanced connectivity and novel applications, privacy and security of the networks and the applications must be ensured. For example, blockchain has the transparent nature of disclosing potentially private information to all participants. There are several solutions to address this risk, including ring signatures, zero-knowledge parameters, and coin mixing [36]. Meanwhile, 6G will dramatically improve the performance and service of wireless networks, leading to a dramatic increase in ubiquitous computing resources. Blockchain's network layer will benefit from improved performance, as well as the expected increased configurability of wireless networks [37]. For example, Feng *et al.* [38] proposed an efficient and secure 5G data-sharing model based on blockchain to deal with the problems that an open and untrusted environment may bring to authentication and data sharing. The model uses blockchain and attribute-based encryption (ABE) to ensure the security of instruction problems and data sharing. The authentication mechanism in the model adopts a smart contract for authentication and access control.

III. Blockchain-Assisted Software-Defined Content Delivery Network Architecture

There are still many issues to be resolved in the future of software-defined CDNs, one of the most important of which is cross-domain trust. More and more CDN services begin to be under the business scenario of multiparty participation, and the transaction mode will support the dynamic switch of user networks among many operators. However, the uncontrollable factors in the business scenario of multiparty participation increase, and it is difficult to guarantee the credibility of each participant and transaction. For example, users use different operators' network services in a settlement period, and a clear bill is needed for the specific use of network traffic. If the bill is only generated by the operator, users will not be convinced, so a reliable bill generation mechanism is needed to support it. In addition, the relative equality among operators does not allow one party to control the trading rules and process, that is, the traditional architecture is no longer applicable to the application scenario of multiparty participation and equal autonomy.

A. System Model Overview

Blockchain can build trust between different nodes and has the characteristics of peer-to-peer autonomy, data maintenance, and tamper-proof. Therefore, it can be introduced into the software-defined CDN and establish a trusted scheduling mechanism and trust trading patterns with the help of a smart contract.

1) Credible Scheduling Mechanism: Traditional SDN controller face credibility is difficult to ensure that the risk of a single point of failure, instructions, and can use blockchain technology, the scheduling strategy into smart contracts, deployment in the blockchain system, from blockchain when SDN controller platform after listening to the needs of users, the user needs with operators network parameters as the trigger scheduling policy contract, blockchain directly generated from the decision-making results, to avoid the problem of single-point controller scheduling is not credible.

2) Trust Trading Patterns: The trading mechanism based on the blockchain can be written by the trading rules smart contracts through the access control mechanism to trigger the execution of the contract, so as to control the whole process of trading, the operators to participate in the scene, the scheduling results generated, users and operators directly through the blockchain signed a contract to establish trade relations, when the network operators to provide quality no longer meet the demand of users, a new scheduling results generated, users will be signed directly by blockchain system with new operators to establish trade relations.

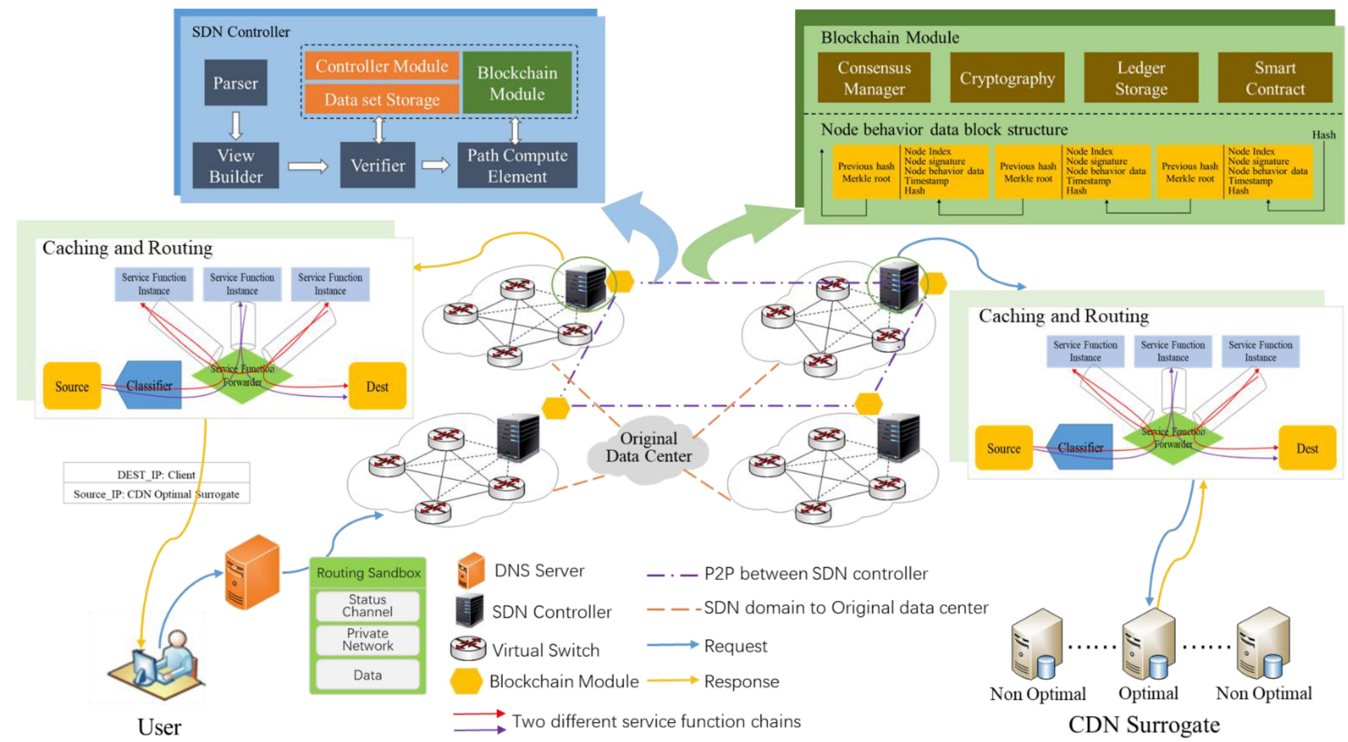

Fig. 1. Blockchain-assisted software-defined CDN architecture.

3) Routing Sanbox: To ensure trusted data delivery, a routing sandbox based on state channel is established. Routing sandboxes can set up private communication paths for single-point communication. Unlike virtual private network mechanisms, verification data for private channels is stored on the blockchain. After the channel is established, packets are transmitted through the intelligent contract-based network management portal, and the credibility check is performed at the exit according to the verification data on the chain.

In addition, blockchain nodes jointly maintain the data stored in the blockchain, which has the advantage of preventing tampering. Once the data goes through the chain of consensus, even if the consensus node exits, it will not cause data loss or failure. Under the content distribution scenario, the decision basis and decision instructions of the control layer will be broadcast and stored in the blockchain system in the form of transactions. The network traffic provided by the operator can be distributed collected through the blockchain as the basis for settlement, and the billing process will also be recorded through the blockchain system. When the two sides of a transaction disagree, the data obtained from the blockchain can be held accountable as evidence.

B. System Components

Based on the credibility analysis in the multiparticipant scenario described earlier, we found that the existing software-defined CDN architecture does not meet the need to ensure that it performs trusted data delivery. At the same time, some existing studies have confirmed that blockchain technology can build trust between different nodes and has the characteristics of peer-to-peer autonomy, data maintenance, and tamper-proof. Further, we propose a blockchain-assisted software-defined CDN. As shown in Fig. 1, the trusted data delivery process is implemented using a software-defined CDN architecture with blockchain help. The SDN controller acts as a peer node of the blockchain in order to establish trust between different regions and participants and achieve peer autonomy. The SDN controller controls CDN cache devices located in the data plane to achieve flexible service choreography. The two main purposes of this architecture are to enhance the security of network communication and to establish trust relationships for entities in different domains. Two main components of the blockchain-assisted software-defined CDN are described as follows.

1) Blockchain Module for Connected Cross-Domain SDN Controller: The blockchain module is mainly responsible for ensuring the credibility of cross-domain services. It consists of four assembly components: 1) consensus management; 2) cryptography; 3) ledger storage; and 4) smart contract. They are described in detail as follows.

Consensus Management: The consensus mechanism is a blockchain governance system, a set of methods designed by combining economics, game theory, and other disciplines to ensure that each node in the blockchain can actively maintain the blockchain system. It was first proposed by Satoshi Nakamoto in the White paper of Bitcoin and gradually developed into an important mechanism to maintain the multicentralization of distributed ledger, which is the core to maintain the safe and stable operation of the blockchain. To be specific, the so-called "consensus mechanism" is to complete the verification and confirmation of transactions within a very short period of time through the voting of special nodes. For a transaction, if several nodes with unrelated interests can reach a consensus. Considering the performance and security issues, we proposed an architecture compatible with mainstream consensus mechanisms, such as PoW, PoS, DPoS, PBFT, and Raft to meet different service requirements. In this article, we adopt the optimized PBFT algorithm, which is introduced in Section IV in detail.

Cryptography: The encryption mechanism is the foundation of the whole blockchain. We will only briefly introduce the encryption mechanism used in the blockchain.

1) The hash function is a kind of mathematical function, which can compress a message of arbitrary length into a binary string of fixed length within a reasonable time, and its output value becomes a hash value, also known as a hash value. The hash algorithm is often used to realize data integrity (message digest) and entity authentication (signature), and it also constitutes a security guarantee for various cryptographic systems and protocols.
2) The elliptic curve cryptography (ECC) algorithm is an asymmetric encryption algorithm based on elliptic curve mathematics. Its security depends on the difficulty of offline logarithm of elliptic curve.
3) *Zero-Knowledge Proof:* The verifier can make the verifier believe that they have the knowledge without telling the verifier what the knowledge is.

Ledger Storage: In our proposed architecture, network state information is contained in a public ledger in the form of blocks, as shown in Fig. 1 of the node behavior data block structure, with each block anchored to the blockchain. The ledger is made up of blocks linked together by a block head and a block body, which mainly stores transaction information. The block head is used to connect the blocks and form a chain. Data are stored as a Merkle tree, and its root is generated by a hash algorithm. Any modification behavior can change the Merkle root value, which means that node behavior can be tracked and not tampered with. The node behavior data block contains important information that can be analyzed, such as node number, node digital signature, a large amount of node behavior data, and timestamp.

Smart Contracts: Smart contracts are event-driven, multirecognized programs that run on the blockchain and automatically process services based on preconditions. Smart contracts, once written, can be trusted by users and cannot be changed. Therefore, smart contracts can enable blockchain technology to be applied to the practical application plane.

2) SDN Controller Structure for CDN Point of Presence: Fig. 1 also shows the SDN controller structure in the blockchain-assisted software-defined CDN architecture. The SDN controller consists of Parser, View Builder, Verifier, Path Compute Element, Data Set Storage, Controller Module, and Blockchain Module. Four of the components are described in detail as follows.

Parser: The switch sends Packet_In, Flow_Mod, Stats_Reply, and Features_Reply information to the controller through the southbound interface. Packet Parser parses the data packet sent by the switch. Packet_In is the signaling for the switch to request the flow table from the controller; Flow_Mod is the signaling for the controller to send the flow table after processing the request; Stats_Reply contains the status information of the switch; and Features_Reply is the signal for the switch Function characteristic information, including path identifier (path_id) and flow table information (such as Flow Entry), etc. Packet Parser dynamically monitors and parses the data packets passed into the controller and extracts important metadata sets.

TABLE I
COMPARISON OF CONSENSUS MECHANISMS

Consensus	PoS	DPoS	Casper	PoET	PBFT
Decentralized	full	full	full	full	full
Tokens	yes	yes	yes	no	no
Evil number	51%	51%	51%	51%	33%
Performance	middle	high	middle	high	high

View Builder: Responsible for analyzing metadata sets, extracting switch network topology and status information, and playing an important role in locating network faults and optimizing the network reasonably.

Verifier: From the perspective of system security, in order to prevent attackers from modifying packet data or injecting additional packets to damage the system security, it is necessary to verify the legitimacy of the node.

Path Compute Element: The path compute module in the SDN controller is used to calculate the optimal path according to the relevant configuration information of the service. The calculation of the optimal path needs to consider some factors, such as delay, packet loss rate, bandwidth, jitter, and cost, as well as the weight of the above factors. The routing path from the user source to the optimal CDN surrogate can be calculated based on the composite measurement of each link, and then sent to the SDN switch through the flow table. In this article, we focus on bandwidth and delay and achieve dynamic adjustment of weights according to actual network requirements through MPC. This will be described in detail in Section V.

IV. BLOCKCHAIN CONSENSUS MECHANISM

In this section, the blockchain consensus mechanism is introduced, and we propose an optimized BPFT consensus process.

A. Blockchain Consensus

In the blockchain asynchronous network, state replication between hosts is necessary to ensure a consistent state for each host. The consensus algorithm ensures consistency between hosts after state replication, which is the most fundamental and important part of the blockchain. Based on the consortium blockchain, a variety of consensus algorithms have been designed as shown in Table I, such as proof of work, PoS, DPoS, Casper, Proof of Elapsed Time (PoET), and PBFT. Later, we will detail the PBFT consensus process and the enhancements we have made to it.

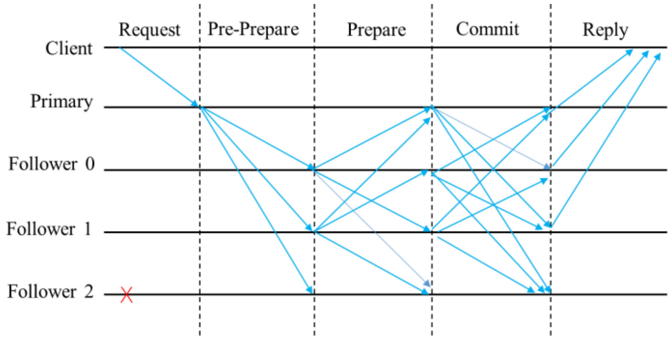

Fig. 2. Schematic of PBFT consensus process.

B. Optimized PBFT Consensus Mechanism: TS-PBFT

Aiming at the shortcomings of the PBFT algorithm directly applied to the SDN control plane, such as low consensus efficiency and poor self-healing, this article proposes a simplified trust-based PBFT algorithm, introduces a trust evaluation mechanism, uses node reputation as an election proof, and combines it with the simplified PBFT algorithm. Achieve a virtuous circle of the system, reduce communication costs, and improve efficiency. At the same time, this article also enhances system security by limiting communication time and adding intrusion detection mechanisms.

There are primary nodes (Primary) and regular nodes (follower) in the protocol. The primary node mainly has the following three functions.

1) When working normally, receive the transaction request from the client, set the number for the request after verifying the identity of the request, and broadcast the preprepare message.
2) When the new primary is elected, it sends View-News messages according to the View-Change messages collected by itself to let other nodes synchronize data.
3) Primary maintains a heartbeat with other nodes.

It should be noted that the status of the primary node is the same as that of the follower node, and it has no privileges. If it is down, no messages occur, messages with incorrect numbers or tampered messages will be sensed by other nodes and trigger view-change.

1) Traditional PBFT: In an SDN control plane consensus process, there is only one master node, which is responsible for verifying the transactions received within a period of time, and the verified transactions will be packaged into the block. The nodes participating in the consensus will have different numbers from 0 to $R = 3f + 1$, and the main node selection formula is, where p is the selected node number; v is the view number gradually increasing from zero; and $|R|$ is the total number of nodes participating in the consensus.

As shown in Fig. 2, a complete and detailed consensus process includes the following steps.

1) The system elects a primary in turn according to the number, and the primary initiates a view-new message to synchronize the data of all nodes;
2) The client initiates a request and forwards it to the primary. After the primary is verified, it broadcasts the request, initiates a preprepare message to all follower nodes, and saves the request itself.
3) After all followers receive the preprepare message, the first step is to verify, including data sequence, and transaction signatures (to prevent client fraud or primary node tampering).
4) After the follower verifies correctly, write it to its own disk, then broadcast the Prepare message, and enter the Prepare stage by itself.
5) All nodes count the prepare message for a certain Request. When the statistics result exceeds 2f nodes, it indicates that most of the nodes have completed the persistence, and then enter the commit phase.
6) The node broadcasts the commit message and counts the number of received commit messages. When all the nodes that exceed 2f send commit messages, the node completes the commit phase, writes data, and updates the controller table data.

2) TS-PBFT: Because the execution priority mechanism in the SDN network does not require strict ordering of requests, in this article, the PBFT consensus algorithm is simplified, the trust evaluation mechanism is introduced, and the controller with the highest trust is selected as the primary node to lead the invasion. For detection and security policies, other nodes still act as followers. The simplified consensus process cancels the preprepare process. As shown in Fig. 3, a consensus process includes the following steps.

1) The system elects the primary node according to the trust evaluation mechanism.
2) Any controller node (A) that wants to request broadcasts a request <REQUEST, o, t, c> to other controllers, o: the specific operation requested, operation; t: the timestamp added to the requested information; and c: the controller Node ID REQUEST contains message content and message digest (used for signature verification).
3) Other controllers (follower) verify the source of the request, and if the verification is successful, broadcast a prepare message and enter the prepare phase. If the verification fails, the log information is reported to the primary node to perform intrusion detection and security policies.

The following steps are the same as steps 5 and 6 of PBFT. The controller waits for the reply result <REPLY, v, t, c, i >, enters the Commit phase and completes the consensus process, where v: current view; t: timestamp; c: the number of controllers that initiated the request; i: the number of current reply controllers; and REPLY: the operation result, which must have the signature of the controller i to prevent network interception and fraud.

C. Evaluation of the Reliability of the Controller

This article uses the fuzzy comprehensive evaluation model to analyze the trust attributes of the SDN controller. It mainly includes the following steps.

1) Establish Node Trust Evaluation Factor Set: The factor set is an ordinary set composed of various factors that affect the trust degree of the evaluation node, $U = (u_1, u_2, \ldots, u_m)$,

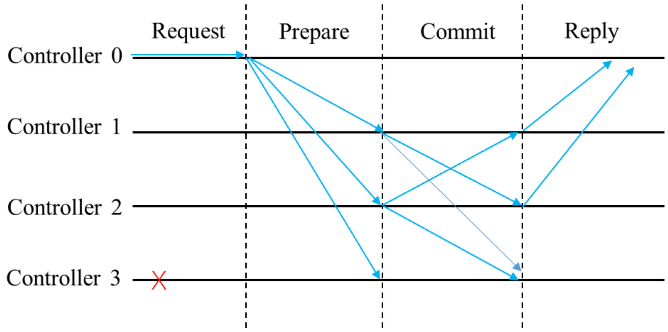

Fig. 3. Schematic of TS-PBFT consensus process.

TABLE II
FUZZY EVALUATION FACTOR SET

Network location	*Nl*	Process or capacity	*Pc*	Duplicate packet ratio	*Rp*	Cache utilization	*Cu*
Link bandwidth	*Lb*	Reach ratio	*Tc*	Flow meter utilization	*Fu*	Delay performance	*Dp*

where u_i is the ith factor that affects the trust degree of the evaluation node. These factors usually have varying degrees of ambiguity. This article chooses the trust evaluation attributes from the perspective of reliability and security.

In terms of security, in order to prevent attackers from modifying data packets or injecting additional data packets to damage the system security, it is necessary to ensure that the received data is reliable, so it is necessary to determine whether the node is legitimate. This article uses blockchain to achieve node identity authentication as shown in Fig. 1. The blockchain uses a public key–private key pair to identify the node's identity. Node identification key pair (PKi, SKi): The public key–private key pair is the identity certificate of node i, SKi is used to sign data packets, and PKi is used for verification. Before forwarding tasks, first, verify the node's identity information. If the verification is successful, the received data will be forwarded. If the verification fails, it is directly determined that the trust level of the node is 0, the node is considered untrusted, and the SDN network is rejected.

After the node is successfully authenticated, analyze the running status of the node from the perspective of reliability, including the following attributes.

In terms of reliability, the above factors are combined with the state and behavior of the node. While ensuring that the node is not malicious, the node must complete the forwarding task quickly and efficiently. The confidentiality and integrity of data cannot guarantee the freshness of data in the network. Incorporating reliability into the trust evaluation system can evaluate node conditions more comprehensively.

2) Establish Evaluation Set for Trust Evaluation: The evaluation set is a set of various results that the evaluator may make to the evaluation object, usually represented by V, $V = (v_1, v_2, \ldots, v_n)$, where the element $v_j$represents the jth evaluation result in Table II. To simplify the process, set judgment $V = \{0, 0.2, 0.4, 0.6, 0.8, 1\}$ is used in this article.

3) Carry Out Single-Factor Fuzzy Evaluation and Obtain Evaluation Matrix: If the membership degree of the ith element in the factor set U to the first element in the evaluation set V is r_{i1}, then the result of the single-factor evaluation of the ith element is expressed as a fuzzy set $R_i = (r_{i1}, r_{i2}, \ldots, r_{in})$, taking m single-factor evaluation sets $R_1, R_2, \ldots, R_m$ as the row. The matrix is called the fuzzy comprehensive evaluation matrix

$$R_{m*n} = \begin{pmatrix} r_{11} & \cdots & r_{1n} \\ \vdots & \ddots & \vdots \\ r_{m1} & \cdots & r_{mn} \end{pmatrix}. \tag{1}$$

4) Determine the Full Vector of Factors: In trust evaluation work, the importance of each factor is different. For this reason, give each factor u_i a weight a_i. The fuzzy set of the weight set of each factor is represented by $A = (a_1, a_2, \ldots, a_m)$, which can be constructed by the pairwise comparison matrix of AHP Weight vector.

5) Establish Trust Evaluation Model: After determining the single-factor evaluation matrix R and the factor weight vector A, the fuzzy vector A on U is changed to the fuzzy vector B on V through fuzzy change

$$B = A_{1*m} \circ R_{m*n} = (a_1, a_2, \ldots a_m) \circ \begin{pmatrix} r_{11} & \cdots & r_{1n} \\ \vdots & \ddots & \vdots \\ r_{m1} & \cdots & r_{mn} \end{pmatrix}$$
$$= (b_1, b_2, \ldots, b_n). \tag{2}$$

It is worth noting that $\circ$ is called the comprehensive evaluation synthesis operator. In order to simplify the calculation, general matrix multiplication is used in this article.

6) Determine the System Score: In order to facilitate the Consortium blockchain to evaluate the trust of nodes in the consensus process, S is defined as the mapping of the response factor scores in V, so the total score of node trust evaluation F can be obtained as

$$F = B_{1*n} * S^T_{1*n}. \tag{3}$$

V. Path Compute Element to the Optimal CDN Surrogate

This article uses MPC to solve the problem of simultaneous optimization of server and path selection in software-defined content distribution networks. Define the system state at time t as Z_t, N_t is the total number of requests in the system at time t, the system input of MPC is $p_{l,s}(t)$, CDN proxy server set $S = \{1, 2, \ldots s, \ldots\}$, and path set $L = \{1, 2, \ldots, l, \ldots\}$. Optimization goal: the user's average response time is the smallest; the user's bandwidth satisfaction conversion is the smallest.

The average user response time is the round-trip delay of the user, including the round-trip link delay and the processing delay of the server. Regarding the server as an M/M/1 queuing model, the processing time of the server can be expressed as

$$d_s(t) = \sum_{s\in S} \frac{\upsilon_t p_s(t)}{\upsilon_s - \upsilon_t p_s(t)}. \tag{4}$$

υ_t represent the rate at which requests arrive at time t, υ_s is the processing speed of the server, and $p_s(t)$ is the probability of the task being processed by the proxy server s. In the SDN network, the controller can obtain the global status information of the network, including the link delay. The link delay detected by the controller is expressed by d_l, then the average round-trip delay requested by the user is

$$d(t) = \sum_{s \in S} d_s(t) p_s(t) + \sum_{l \in L} d_l p_{l,s}(t). \tag{5}$$

Among them $p_{l,s}(t)$ is the probability that the task will be processed by the proxy server through the path.

Therefore, the delay optimization function J_a is defined as

$$J_a = \sum_{k=t+1}^{t+T} d(k). \tag{6}$$

Then consider the user's bandwidth satisfaction, the ratio of the bandwidth that the network can actually provide to the bandwidth requested by the user, so the user bandwidth satisfaction offset on the link l is represented by $B_l(t)$, which is defined as follows:

$$B_l(t) = \left[\frac{\sum_{l \in L} p_{l,s}(t) - C_l}{\sum_{l \in L} p_{l,s}(t) R N_t} \right]^+. \tag{7}$$

R is the average size of user requests, $[x]^+$ represents when $x > 0$, the value is itself, when $x \leq 0$, the value is 0, therefore, the optimization function J_b that defines the bandwidth satisfaction offset is

$$J_b = \sum_{k=t+1}^{t+T} p_{l,s}(t) B_l(t). \tag{8}$$

Finally, consider reducing the change value of the system input $P_{l.s}(t)$ to maintain the stability of the system and define the third optimization function J_c as

$$J_c = \sum_{k=t+1}^{t+T} \sum_{s \in S} \sum_{l \in L} \left| p_{l,s}(k) - p_{l,s}(k-1) \right| \tag{9}$$

In order to consider three optimization functions at the same time, define the joint optimization function as $J_a + \omega_b J_b + \omega_c J_c$, where ω_b and ω_c are the weight values of and, respectively, relative to J_a. By solving the following nonlinear programming problem, all the values in each time slot $t + 1 \leq k \leq t + T$ can be calculated as

$$\begin{aligned} \min \quad & J_a + \omega_b J_b + \omega_c J_c \\ \text{s.t.} \quad & \begin{cases} p_s(t) = \sum_{l \in L} p_{l,s}(t) \\ \sum_{l \in L} \sum_{s \in S} p_{l,s}(t) = 1 \\ 0 \leq p_{l,s}(t) \leq 1 \quad \forall l \quad \forall s. \end{cases} \end{aligned} \tag{10}$$

In the simulation part, the performance change of the algorithm is analyzed by changing the weight value, and the importance of the two optimization goals of response time and bandwidth satisfaction is balanced by adjusting the weight value according to the actual network demand.

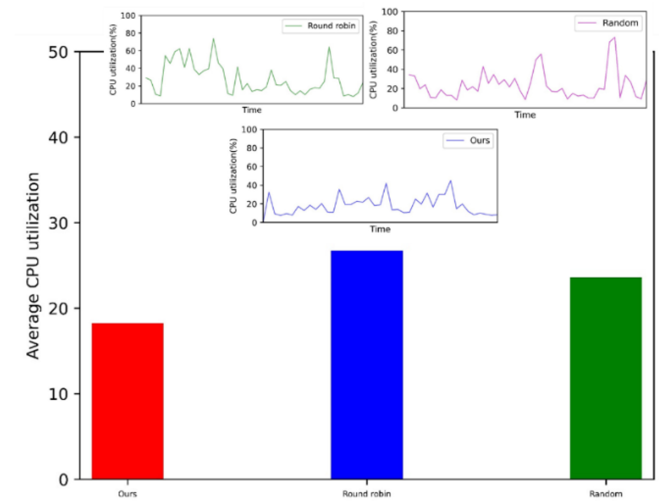

Fig. 4. Comparison of CUP utilization.

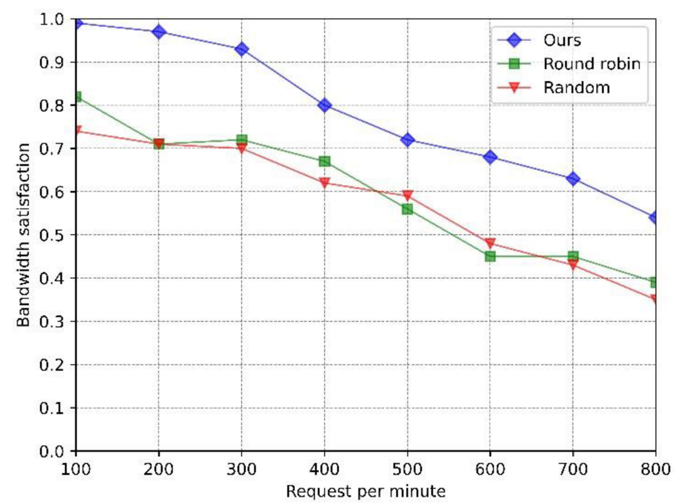

Fig. 5. Comparison of bandwidth satisfaction.

VI. Experiment and Analysis

In this part, we introduce our simulation results and analysis in detail. First, we introduce the open-source simulation tools we use, including SDN virtualization simulation platform Mininet, SDN open-source controller floodlight, global network topology emulator simple Global Network Topology Emulator, and Fast linearization detector Porcupine.

From Fig. 4, we can see that when selecting replica servers in the same network state, our CPU utilization rate is lower than that of round-robin and random, which are 18.25%, 23.59%, and 26.72%, respectively. This indicates that our strategy consumes fewer computing resources while performing the same task. This makes sense to cope with the ever-increasing demand for computing resources created by 6G.

From Fig. 5, we can see that as the user's demand per minute increases, bandwidth satisfaction will decrease, but under the same user request, our bandwidth satisfaction will be higher. However, it should be noted that when the number of requests per minute is less than 300, our performance is more

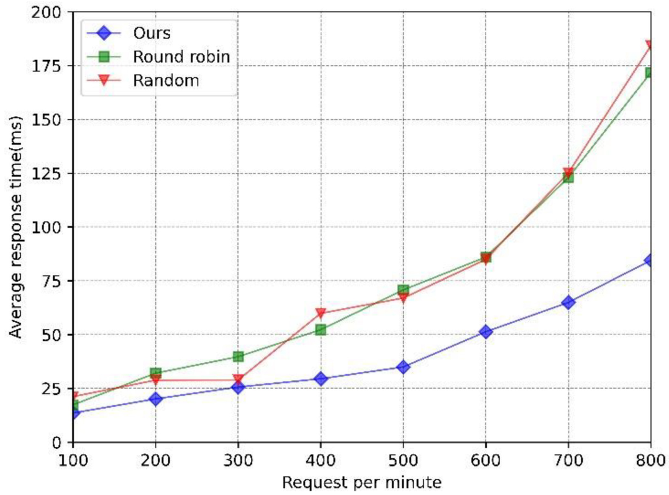

Fig. 6. Response time comparison.

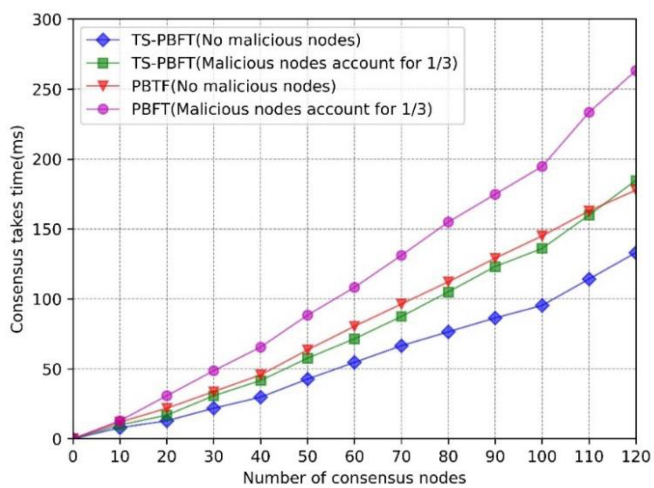

Fig. 7. Comparison of time spent on consensus.

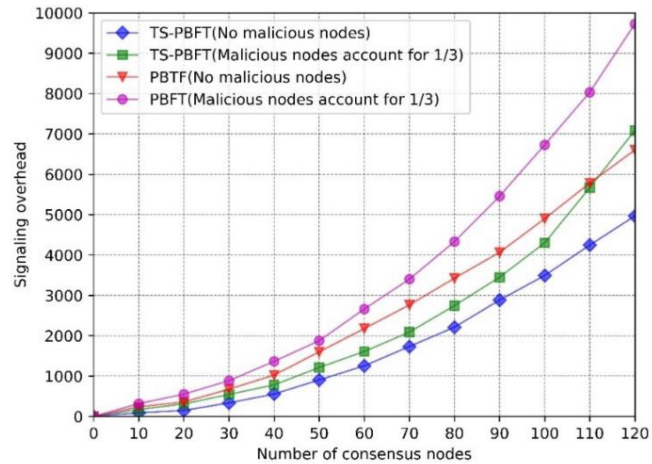

Fig. 8. Comparison of signaling overhead.

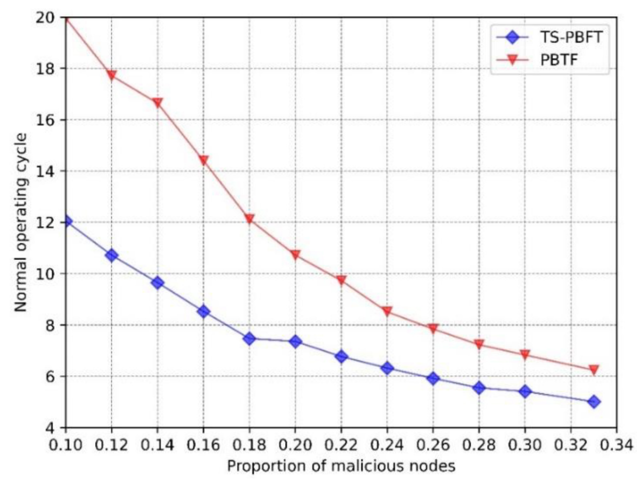

Fig. 9. Comparison of invulnerability.

obvious than that of round-robin and random. However, with the further increase of requests, the downward trend of bandwidth satisfaction in the three methods is similar. The reason is that the importance of response time and bandwidth to meet these two optimization objectives can be adjusted according to the actual network needs to balance the weight values.

The comparison of response time is shown in Fig. 6. As user demand increases per minute, the average response time gradually increases. When the requests per minute are low, our method is more round-robin, and the random advantage is not obvious, but as the requests per minute increase, our method can significantly reduce the average response time.

We compare TS-PBFT and PBFT in terms of consensus spending time, signaling overhead, and resistance in the best (no malicious nodes) and worst (one-third of malicious nodes) network environment. From Fig. 7, we can see that under the same network environment, TS-PBFT spends less time in the consensus process than PBFT; from Fig. 8, it is clear that TS-PBFT has lower signaling overhead than PBFT. The reason is that TS-PBFT reduces a consensus process in the SDN network, and while simplifying PBFT, it can still reach a consensus smoothly. In Fig. 9, we can see that TS-PBFT is not as resistant to defamation as PBFT, but considering the high data volume requirements of software-defined CDNs, TS-PBFT can effectively solve the problem by simplifying the consensus process and reducing signaling overhead. PBFT is directly applied to the problem of low consensus efficiency in SDN scenarios. Considering that TS-PBFT has better performance than PBFT.

We compared the trust evaluation model for normal nodes and two types of malicious nodes. Type A means that malicious nodes directly discard data packets after receiving data; Type B means that malicious nodes repeatedly send the same data packet after receiving data. The result is shown in Fig. 10. Starting from the 12th cycle, the model can distinguish between normal nodes and malicious nodes. Moreover, as the cycle increases, the evaluation of normal nodes will increase and the evaluation of malicious nodes will decrease. Type A nodes have lower scores than type B nodes.

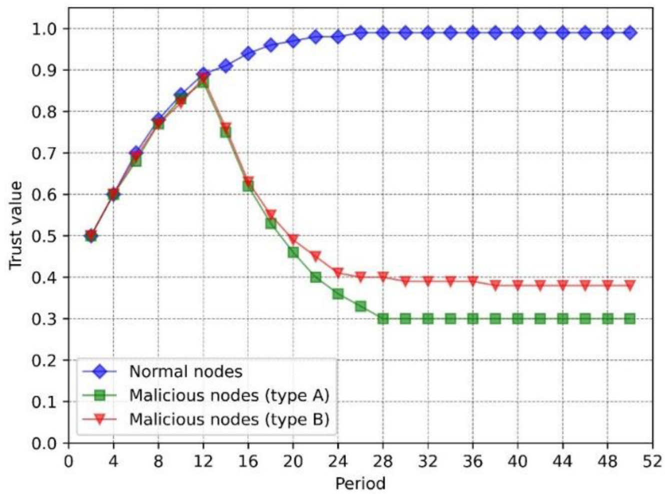

Fig. 10. Trust evaluation in different scenarios.

VII. Conclusion and Future Work

On the one hand, we use the SDN technology to perceive the network topology and provide a global view for the optimal selection of CDN replica servers; on the other hand, the consortium blockchain is built on the SDN control plane where the distributed deployment controllers participate in the consensus process to deal with a single point of failure, malicious node intrusion, and consensus issues. In addition, we use MPC to solve the joint optimization of delay and bandwidth satisfaction in the process of selecting the optimal replica server; establish a fuzzy comprehensive evaluation model to evaluate the trust of the SDN controller to prevent untrusted nodes from joining the SDN network; TS-PBFT is proposed to overcome the shortcomings of low consensus efficiency and poor self-healing when the PBFT consensus algorithm is directly applied in the SDN scenario. Simulation experiments have been done to prove it.

While blockchain technology holds the promise of providing a credible solution in the business scenario of software-defined content distribution networks, there are still some constraints. The constraints are mainly reflected in system stability, business model supportability, application security, and other aspects. Most blockchains cannot meet the requirements of "high efficiency," "decentralization," and "security" at the same time, that is, impossible triangulation. Presently, blockchain is restricted to different degrees in terms of transaction throughput, block capacity, trading capacity, etc., which leads to network congestion and high-frequency business needs that are difficult to be met. Furthermore, problems, such as privacy protection, smart contract vulnerability, consensus mechanism, private key protection, computational force attack, cryptography algorithm security, etc., make the blockchain still face certain challenges in terms of low-level security and application security.

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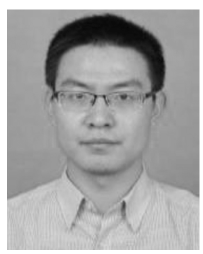

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